

L1

GC Code

e2h2zotc

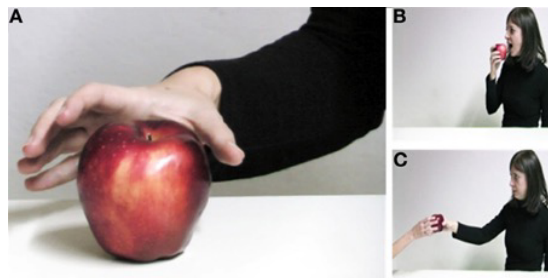
Motor Movement



<https://www.biologyonline.com/tutorials/control-of-body-movement>



<http://nupurnritya.com/courses/degree-courses-in-kathak/>



Becchio et al, 2012



<https://www.morganmckinley.com.au/article/all-important-first-handshake>



<https://www.thestatesman.com/entertainment/farhan-akhtar-croons-telugu-song-for-mahesh-babus-bharat-ane-nenu-1502615016.html>

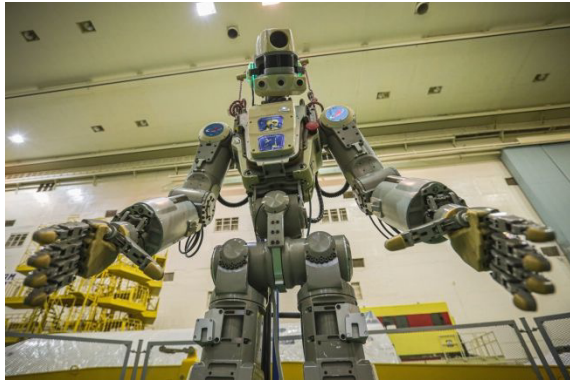
<https://www.musicradar.com/how-to/improve-your-acoustic-folk-guitar-playing-in-20-minutes>

Why study motor cognition?

Why study motor cognition?

Robotics

F.E.D.O.R: Skybot



Sophia



Prosthetics

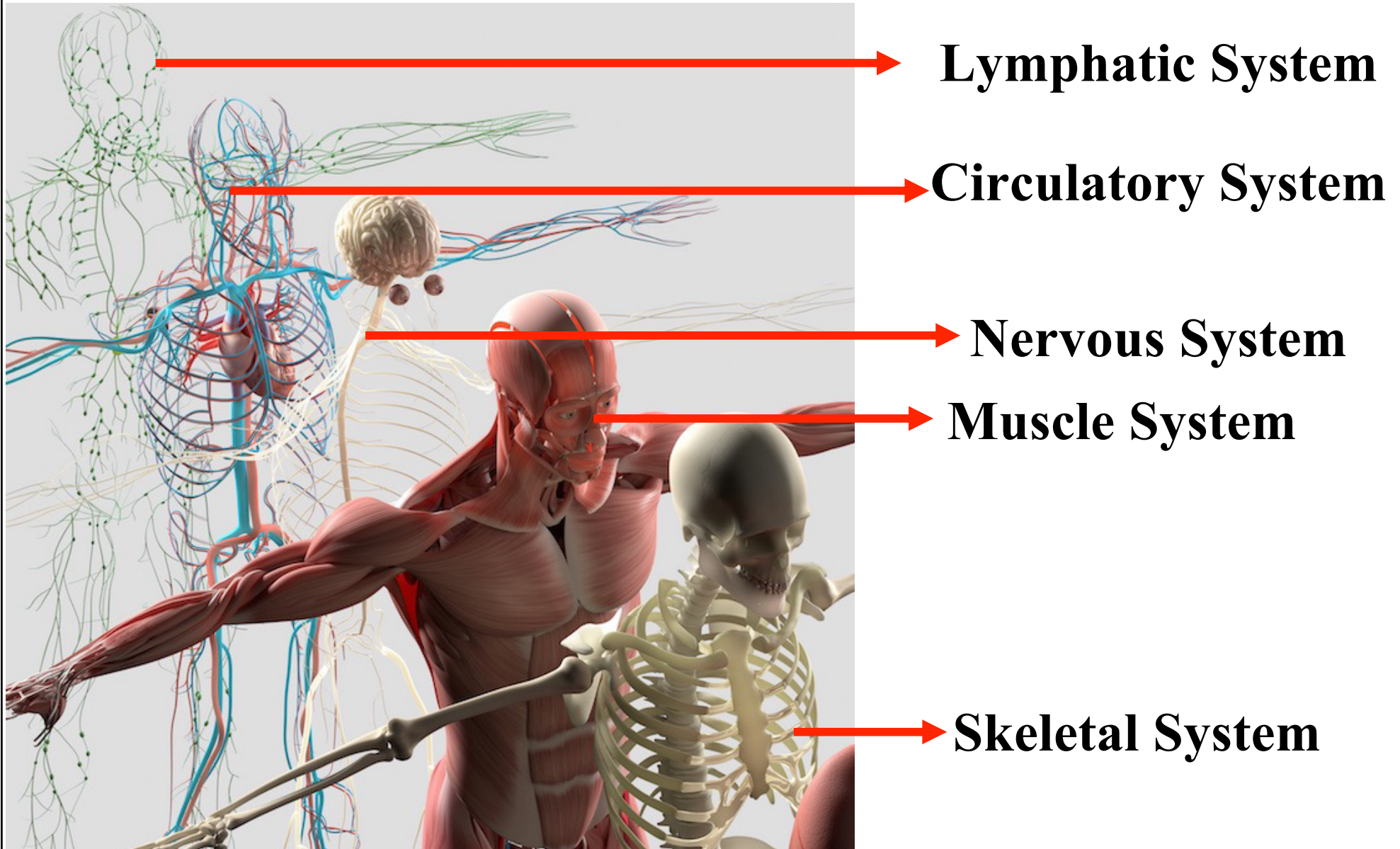


<https://www.sralab.org/center-bionic-medicine/powered-prosthetic-arm>



https://www.thestar.com/sports/2013/02/14/blade_runner_oscar_pistorius_charged_with_murder_in_shooting_of_girlfriend.html

Human Body



The information flow



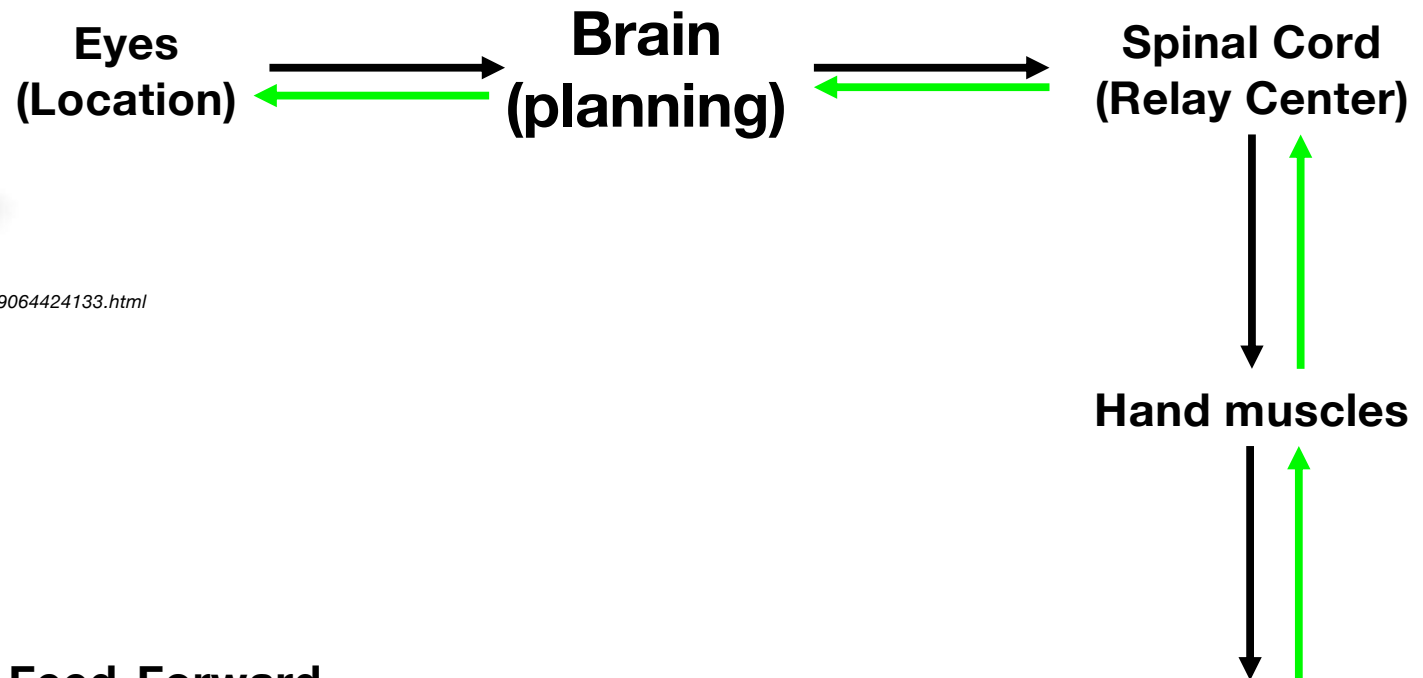
<https://nieuws.kuleuven.be/en/content/2015/how-brain-regions-work-together-to-make-grasping-possible>

- **Eyes**
- **Brain**
- **Arm**
- **Spinal cord**
- **Fingers**

The motor pathway

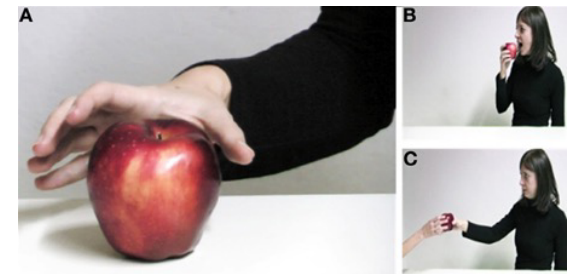


<https://www.indiamart.com/proddetail/fresh-apple-19064424133.html>



→ Feed-Forward

→ Feed-Back

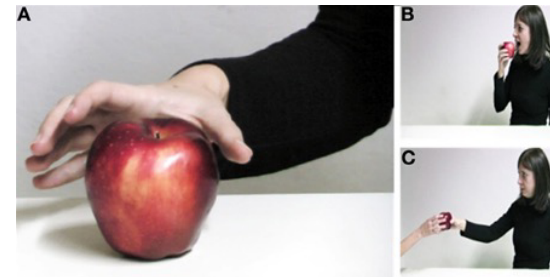
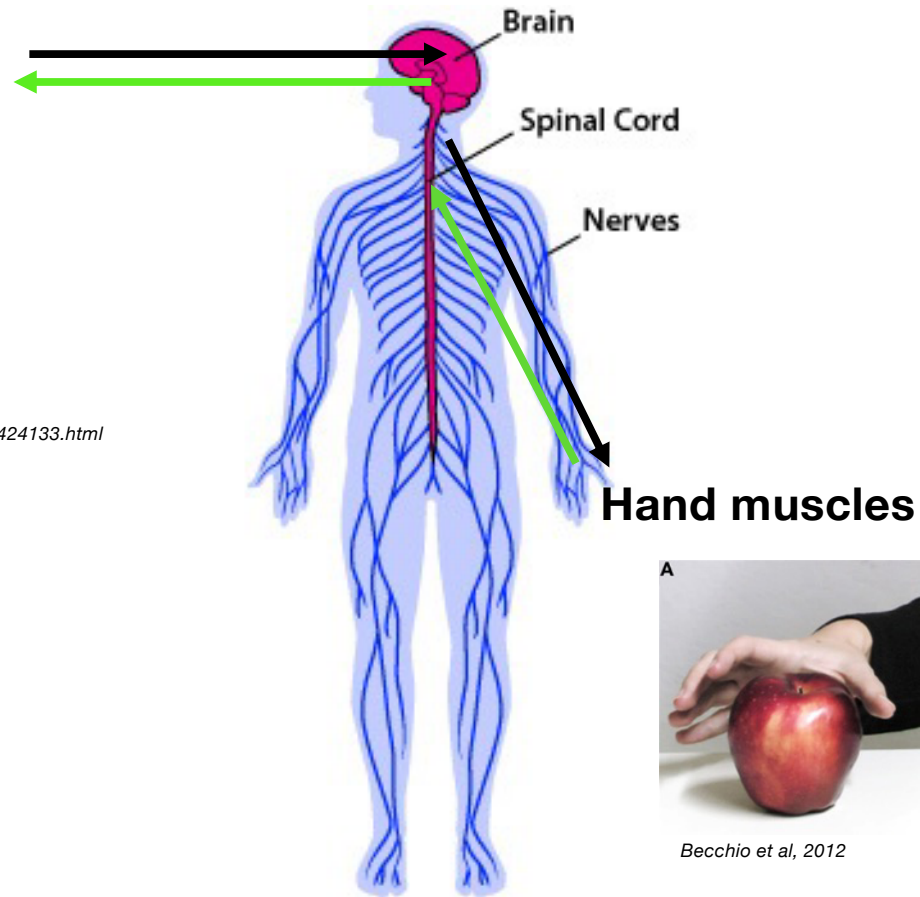


Becchio et al, 2012

The motor pathway



<https://www.indiamart.com/proddetail/fresh-apple-19064424133.html>



Becchio et al, 2012

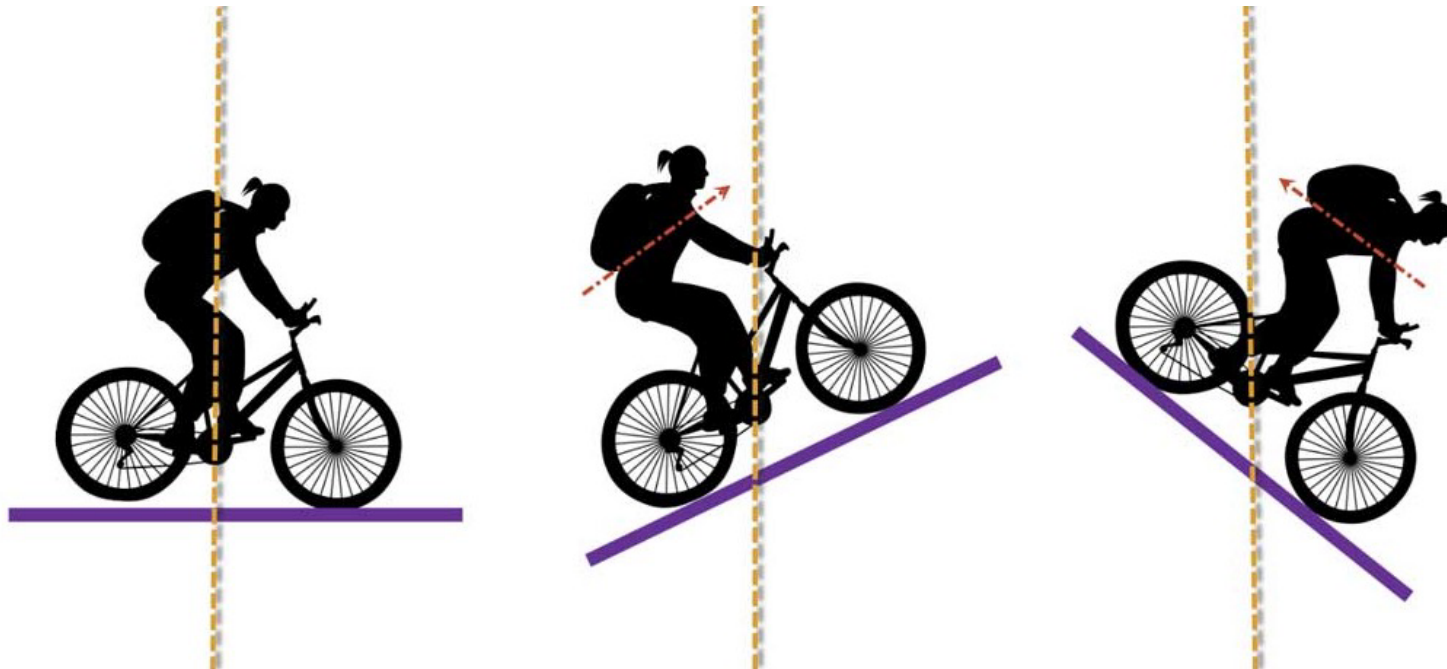
■ Central Nervous System (CNS)
■ Peripheral Nervous System (PNS)

<https://www.google.com/url?sa=i&url=https%3A%2F%2Faskabiologist.asu.edu%2Fparts-of-nervous-system&psig=AOvVaw1iUta9pwlykZDAIP-S746s&ust=1594884992424000&source=images&cd=vfe&ved=0CA0QjhqFwoTCOIKeTtzuocFQAAAAAABAY>

Cycling



<https://www.scientificamerican.com/article/why-dont-we-forget-how-to-ride-a-bike/>



<https://www.liv-cycling.com/global/campaigns/uphill-and-downhill-body-position-on-a-mountain-bike/19049>

Human Motor control

- **Computational:** Mathematical calculations to achieve a movement
- **Procedural:** The behavioural and cognitive states to execute a movement.
- **Implementation:** How muscles contract/functions. The part of brain contributing in execution of a movement.

Interdisciplinary

David Marr Tri-level hypothesis

Brain



<https://medium.com/young-coder/how-illusions-hijack-your-brain-a3dc1a223b03>

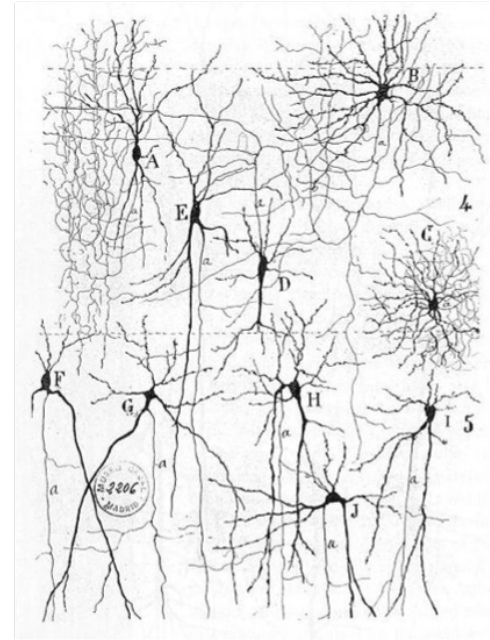
Where is information processed?

Neurons



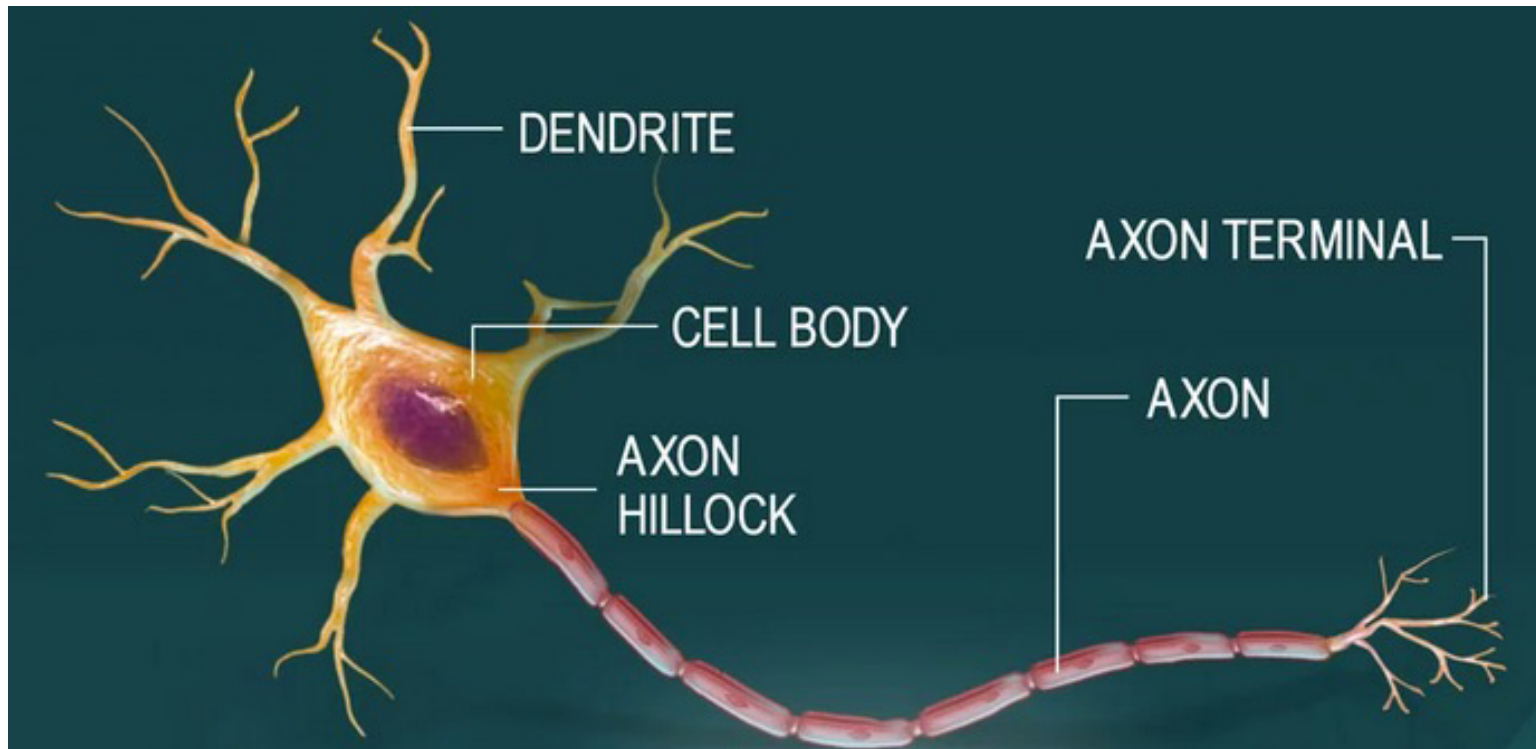
https://www.123rf.com/photo_14119586_human-brain-medical-symbol-represented-by-a-close-up-of-neurons-and-organ-cell-activity-showing-inte.html

Neural Network

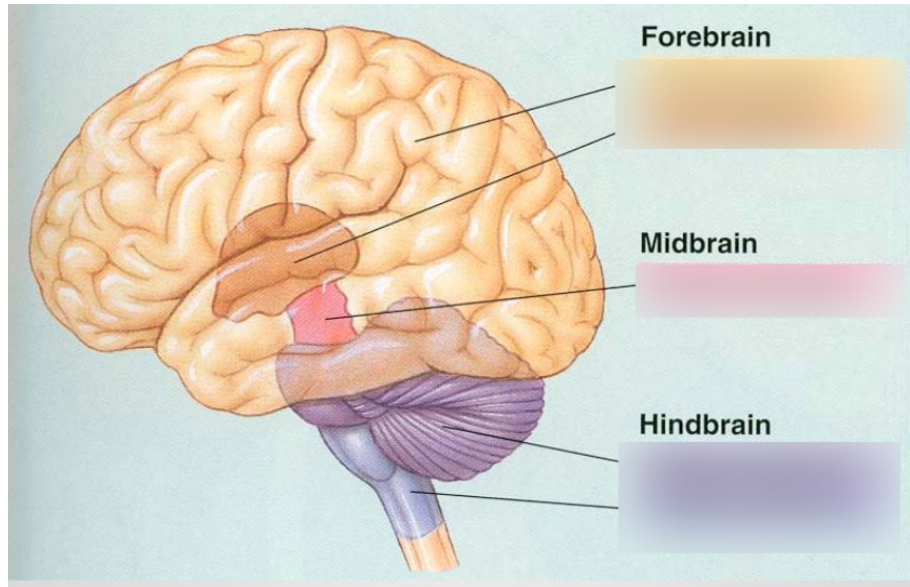


<https://courses.lumenlearning.com/boundless-psychology/chapter/neurons/>

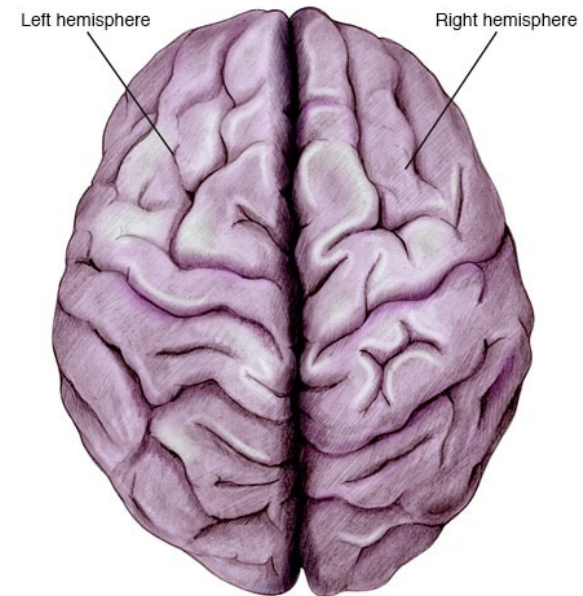
Neurons



Brain



<https://quizlet.com/ca/263942935/forebrain-midbrain-hindbrain-251-diagram-3-diagram/>



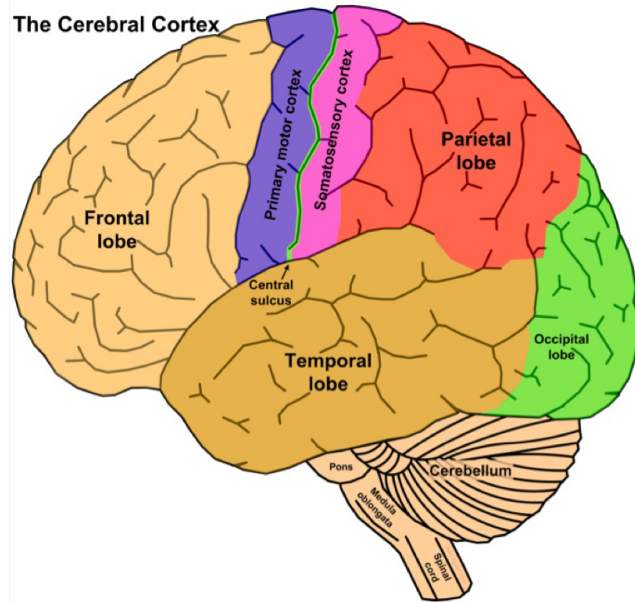
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<https://www.mayoclinic.org/tests-procedures/epilepsy-surgery/multimedia/brain-hemispheres/img-20008029>

- Divided into two parts
 - Right hemisphere
 - Left hemisphere
- The two hemispheres are connected with corpus collosum: Tract of nerve fibers for communication between two hemispheres.

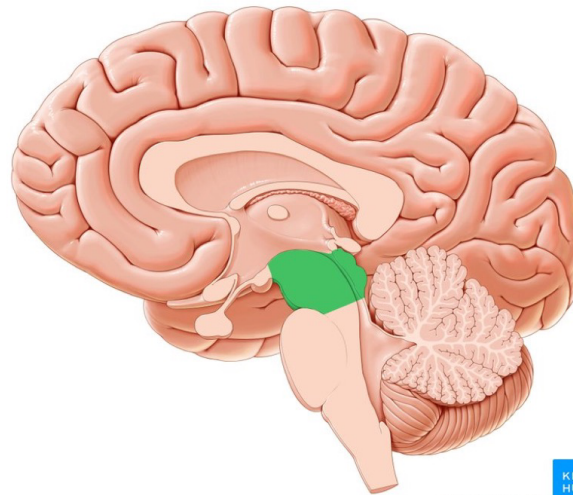
Brain

Forebrain



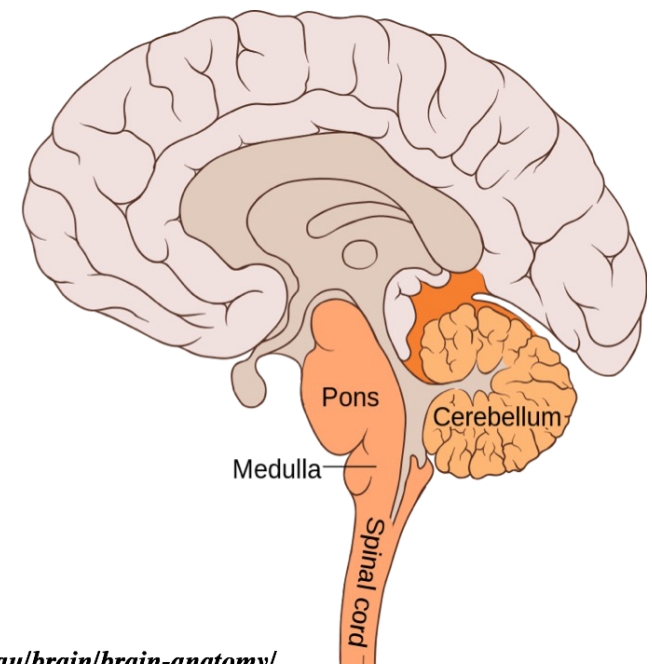
http://www.delib.org/wiki/images/7/76/Cerebral_Cortex.gif

Mid Brain



<https://www.kenhub.com/en/library/anatomy/midbrain-pons-gross-anatomy>

Hind Brain

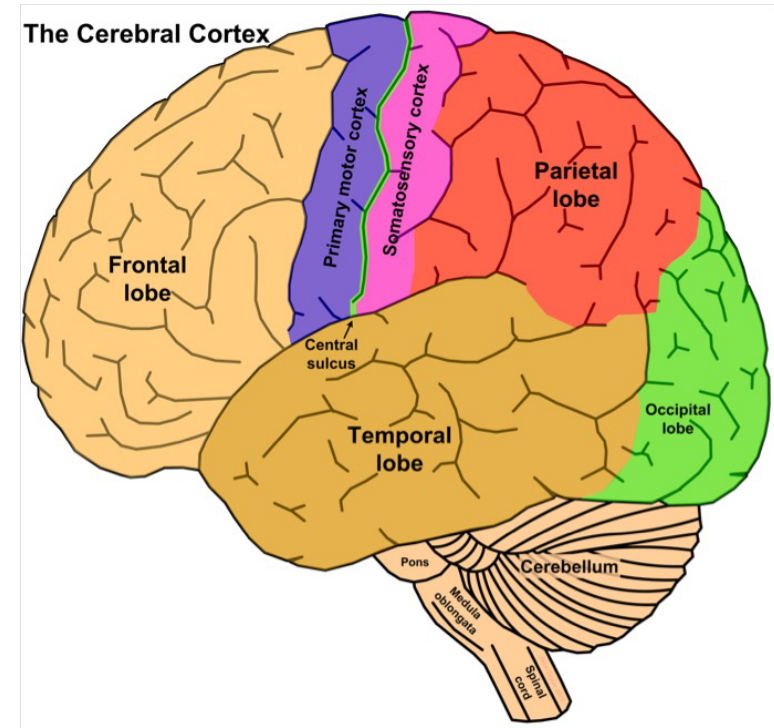


<https://qbi.uq.edu.au/brain/brain-anatomy/hindbrain>

Forebrain

Each hemisphere is divided into four lobes:

- Frontal
- Temporal
- Parietal
- Occipital

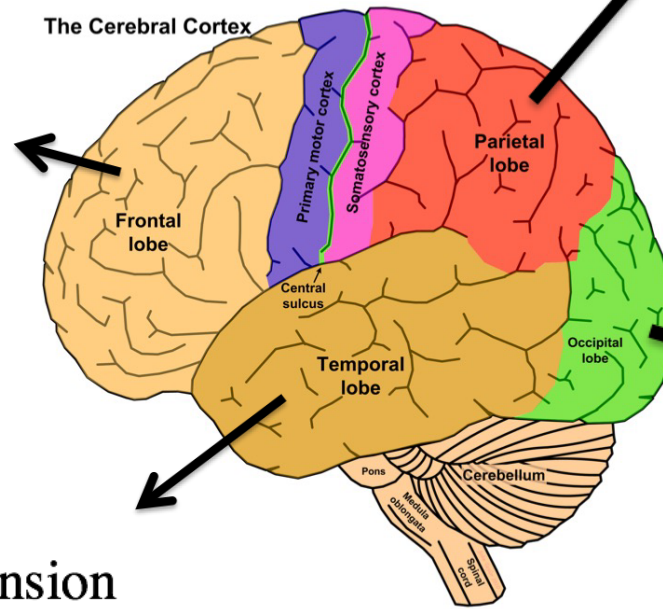


http://www.delib.org/wiki/images/7/76/Cerebral_Cortex.gif

Cerebral Cortex

- Decision making
- Reward
- Speech and Language Control
- Memory formation

- Audition
- Face recognition
- Speech comprehension
- Color perception

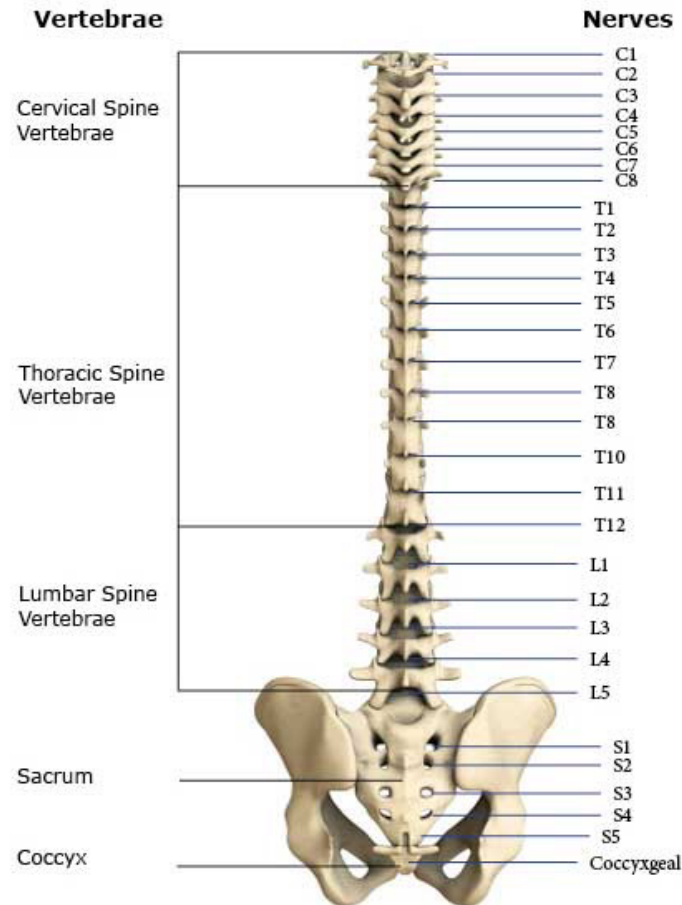


- Attention
- Eye movement
- Heading direction
- Spatial location

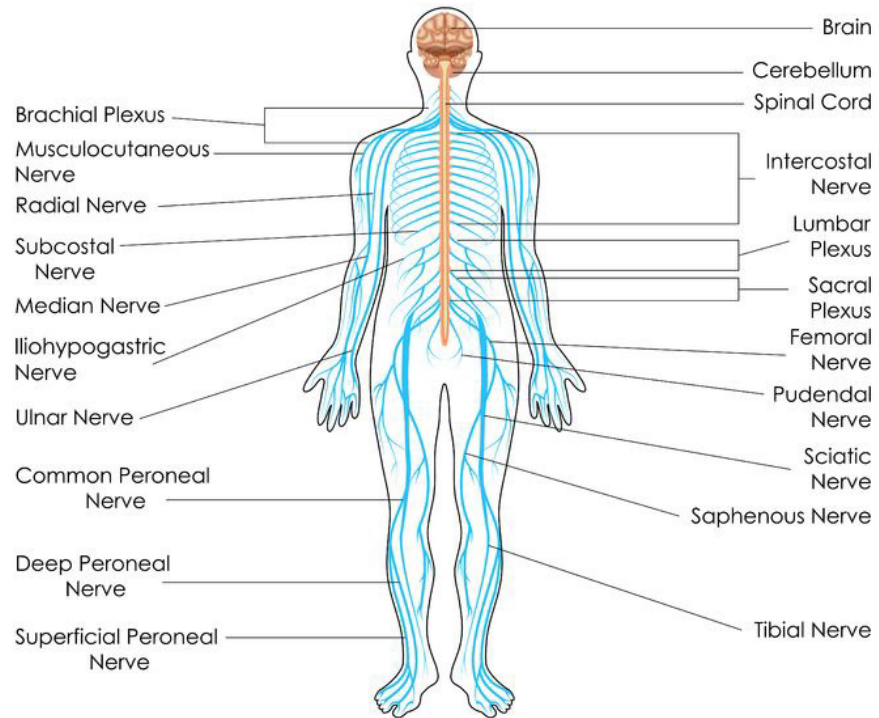
- Vision
- Depth perception

Spinal Nerves

Relation of Spinal Nerves to Vertebrae

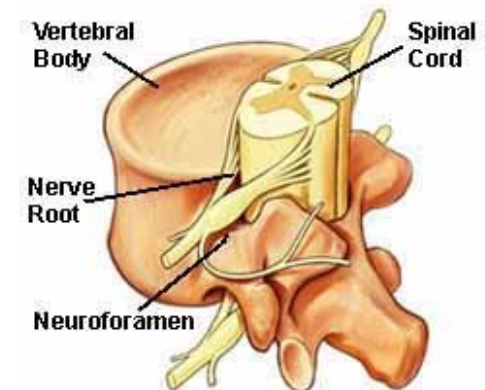


<https://www.healthpages.org/anatomy-function/spinal-nerves/>

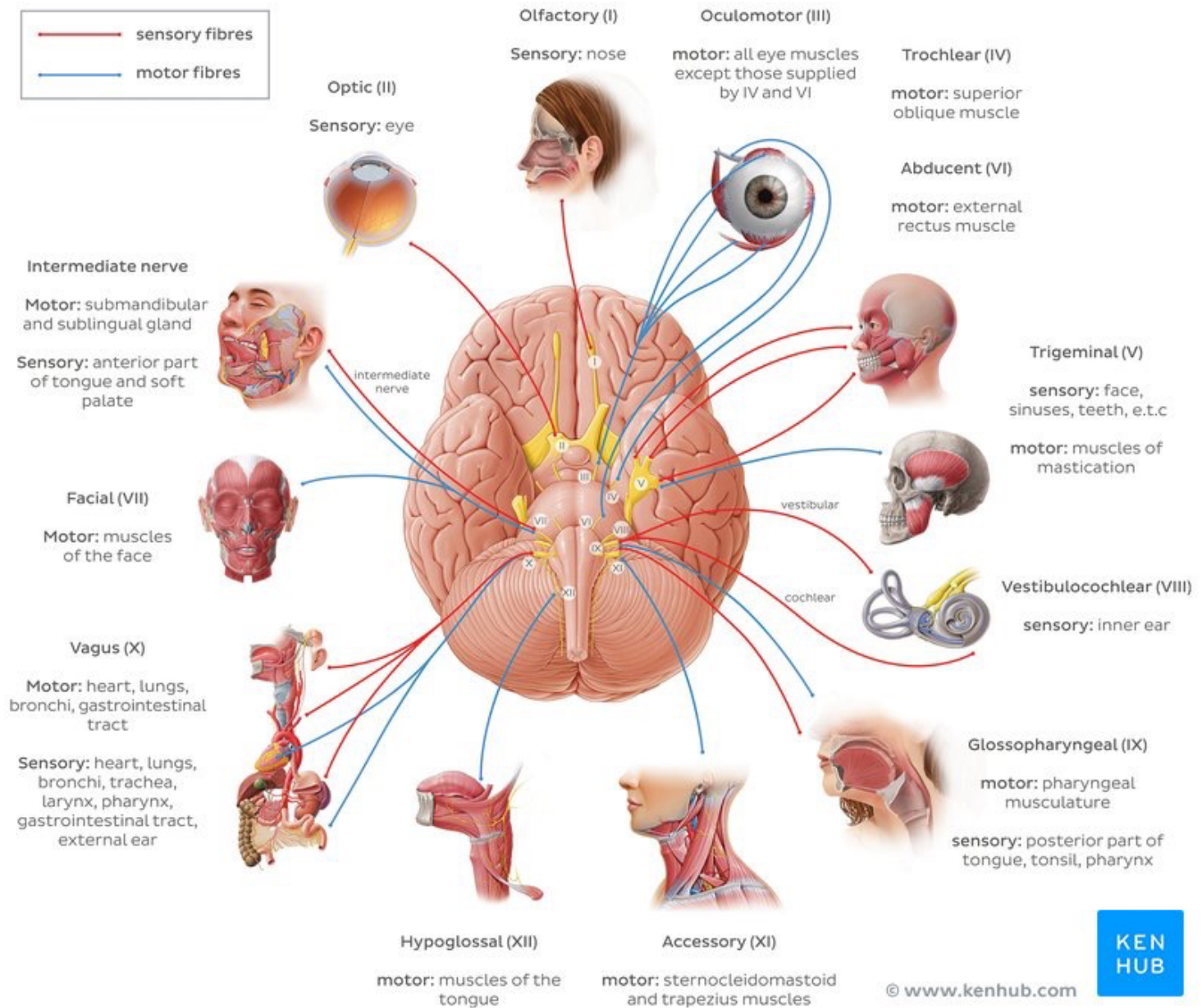


NERVOUS SYSTEM

<https://www.spineuniverse.com/anatomy/nerve-structures-spine>



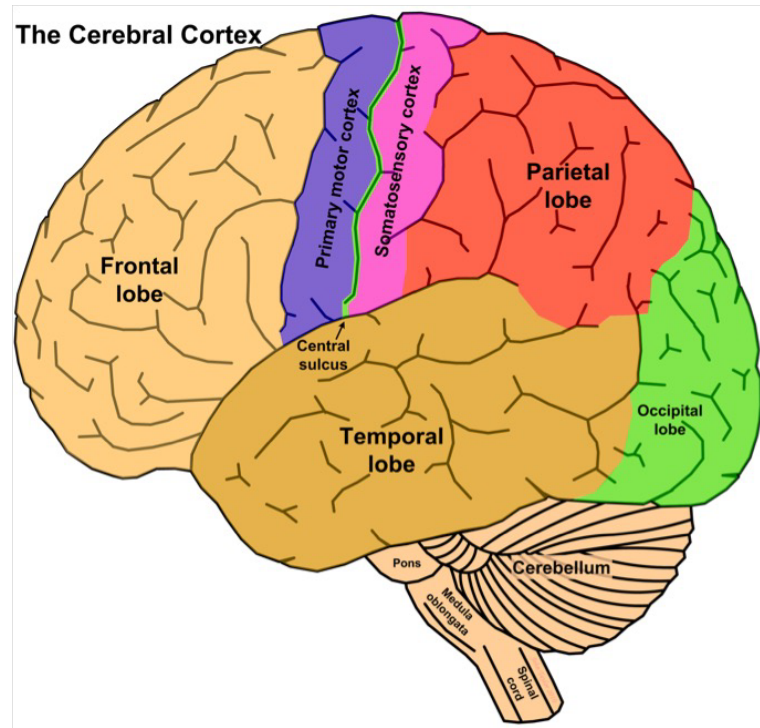
Cranial Nerves



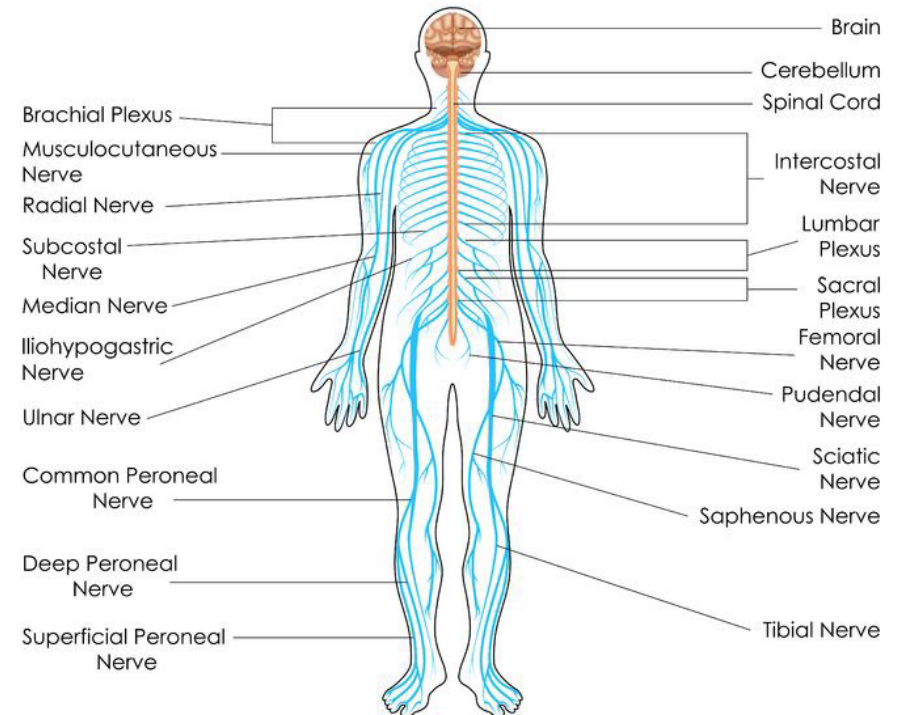
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Information relay

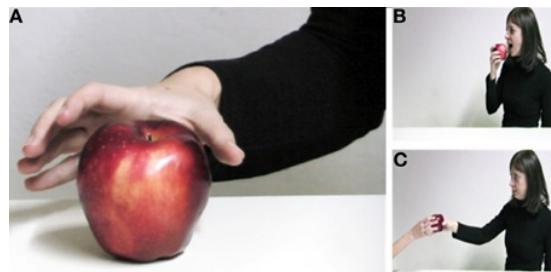


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NERVOUS SYSTEM

<https://www.spineuniverse.com/anatomy/nerve-structures-spine>

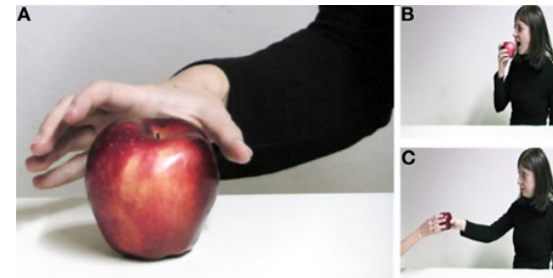
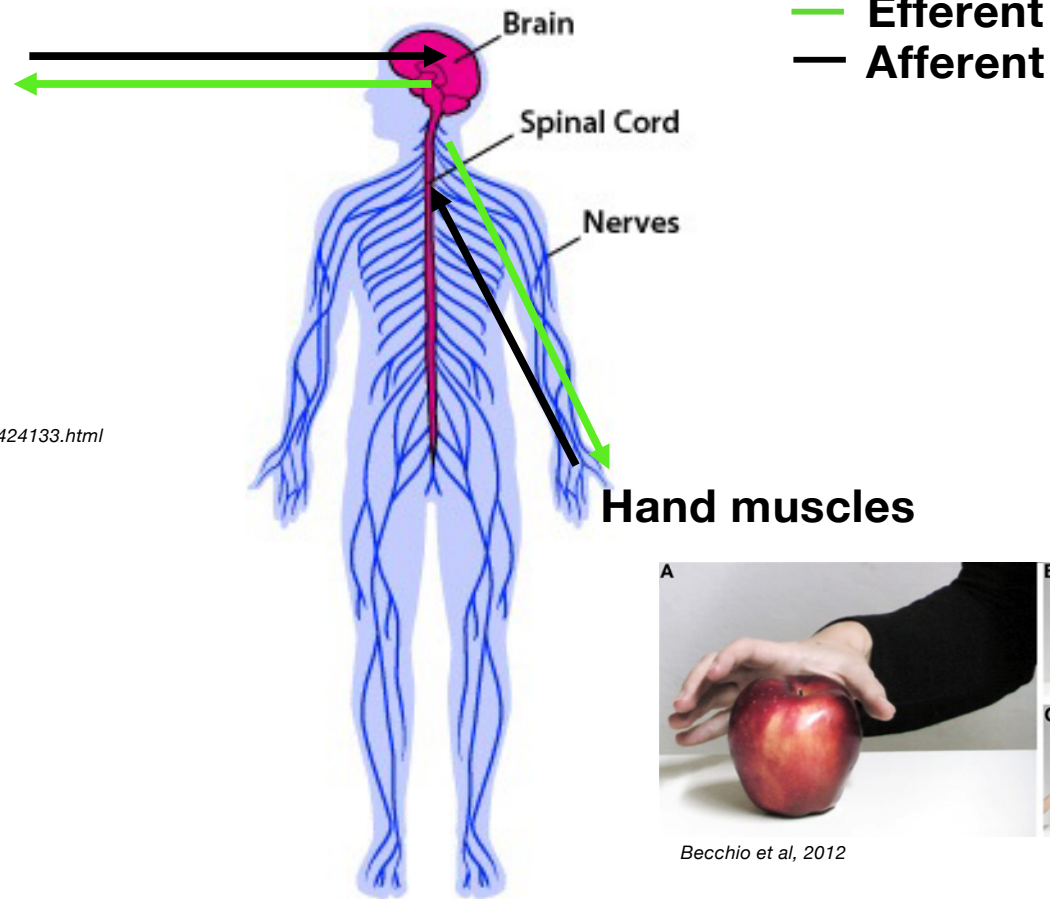


Becchio et al, 2012

The motor pathway



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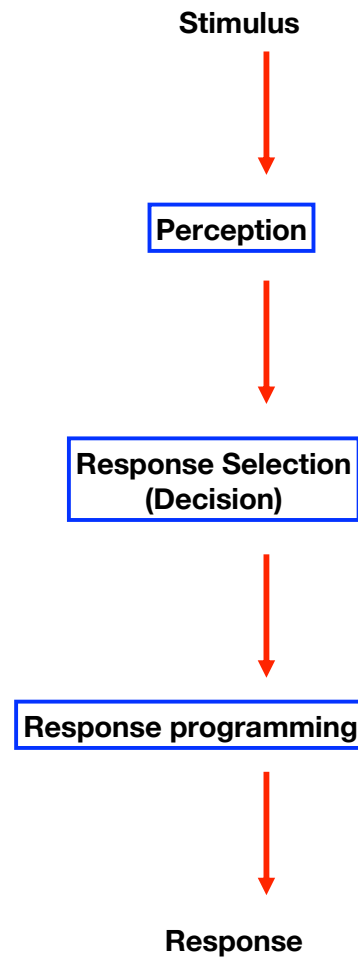


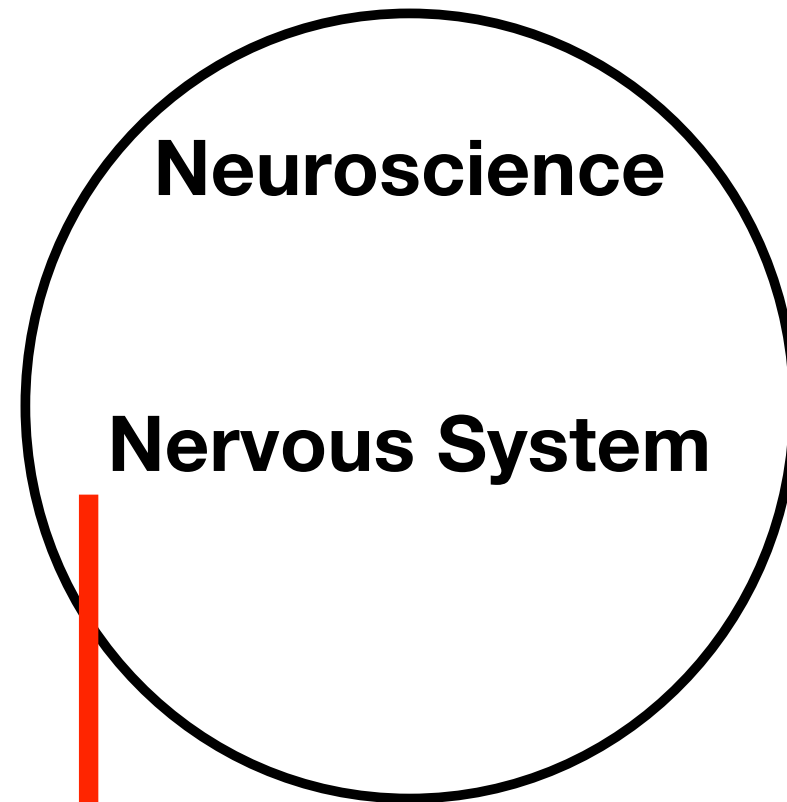
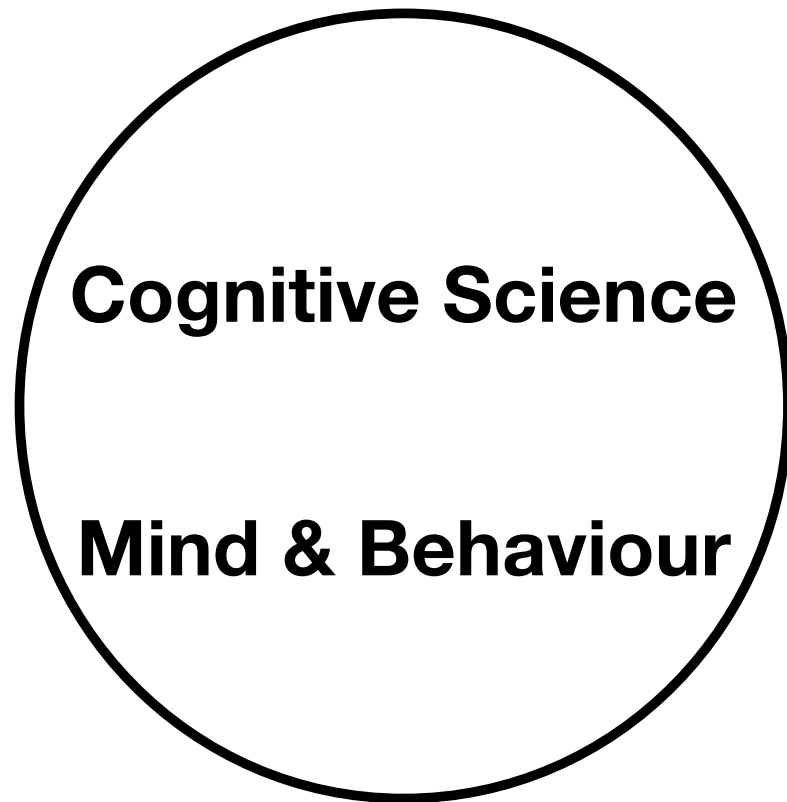
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Information Processing





Cognitive Science

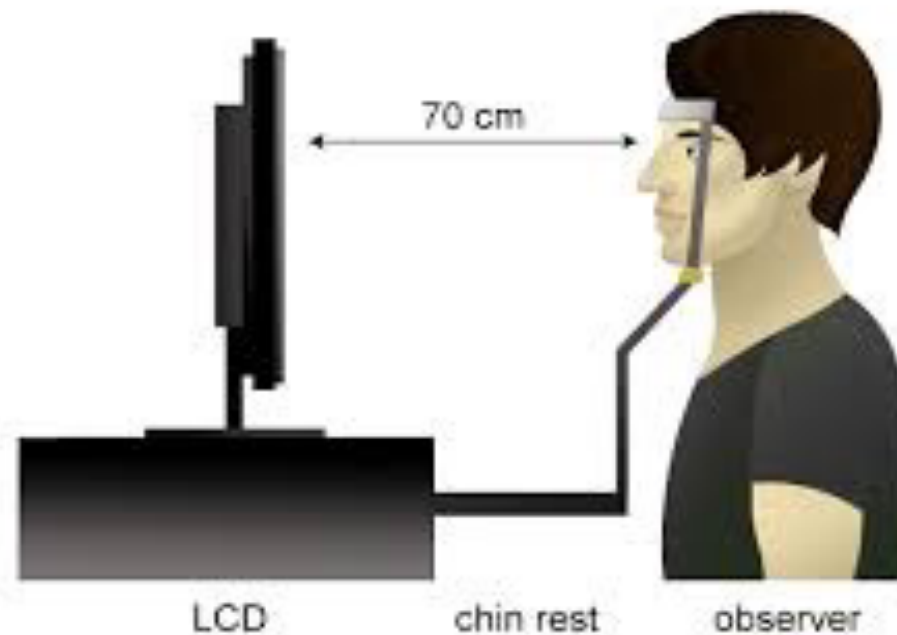
Psychology

Study of Mind → Study of Behaviour → Psychophysics

Technologies



Cognitive Science: Interdisciplinary study of mind and its processes.



Brain

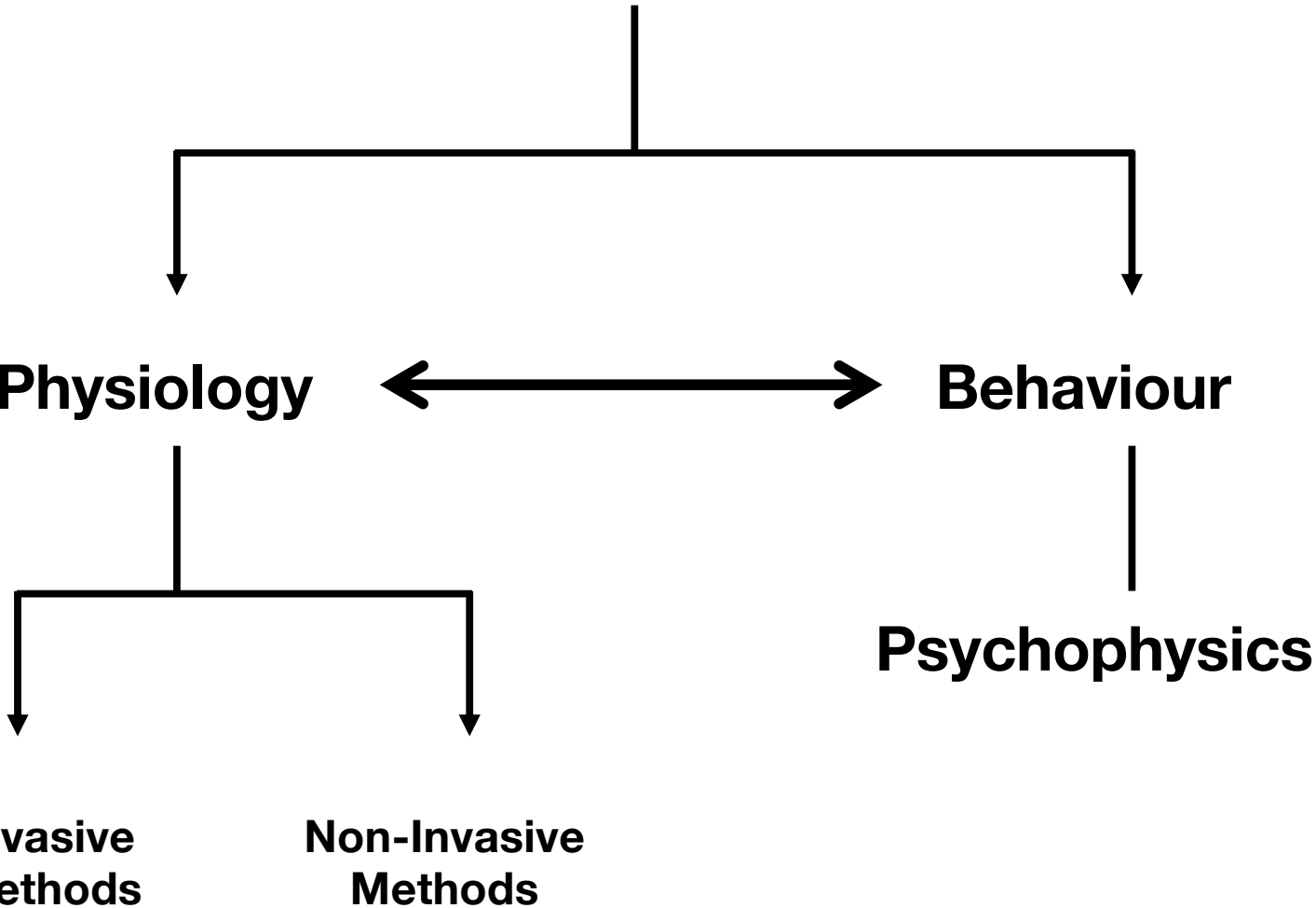
Physiology

Behaviour

Psychophysics

**Invasive
Methods**

**Non-Invasive
Methods**

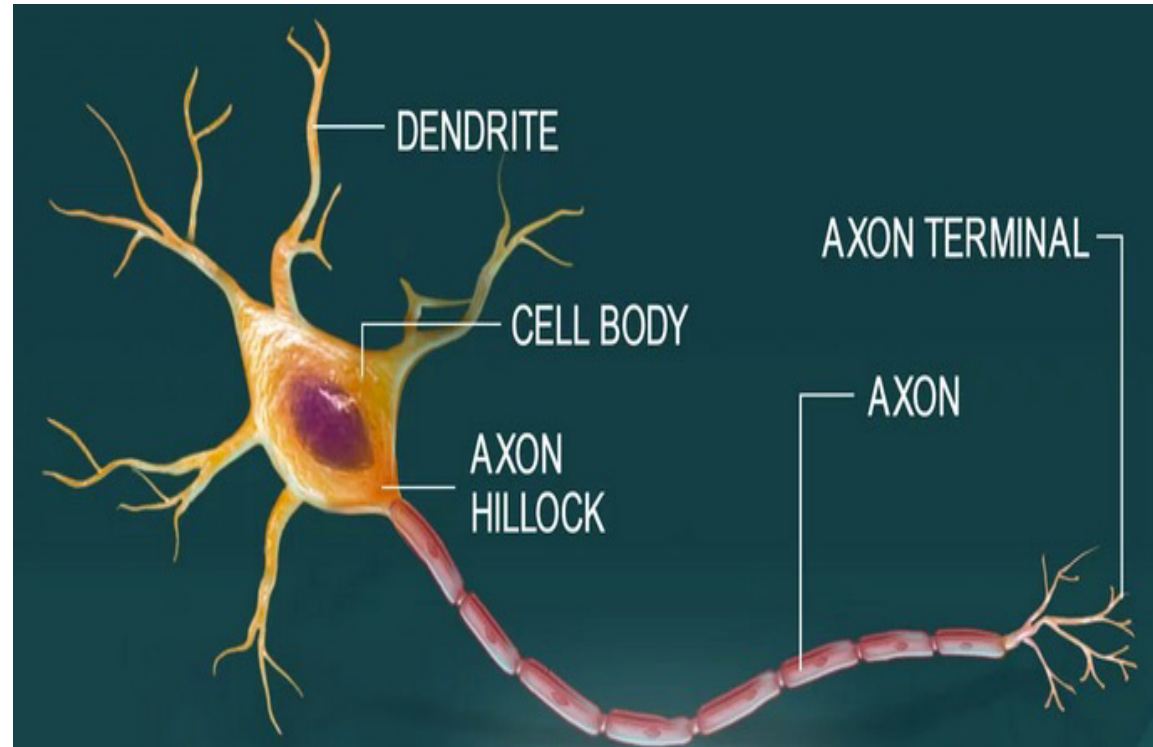


Neurons



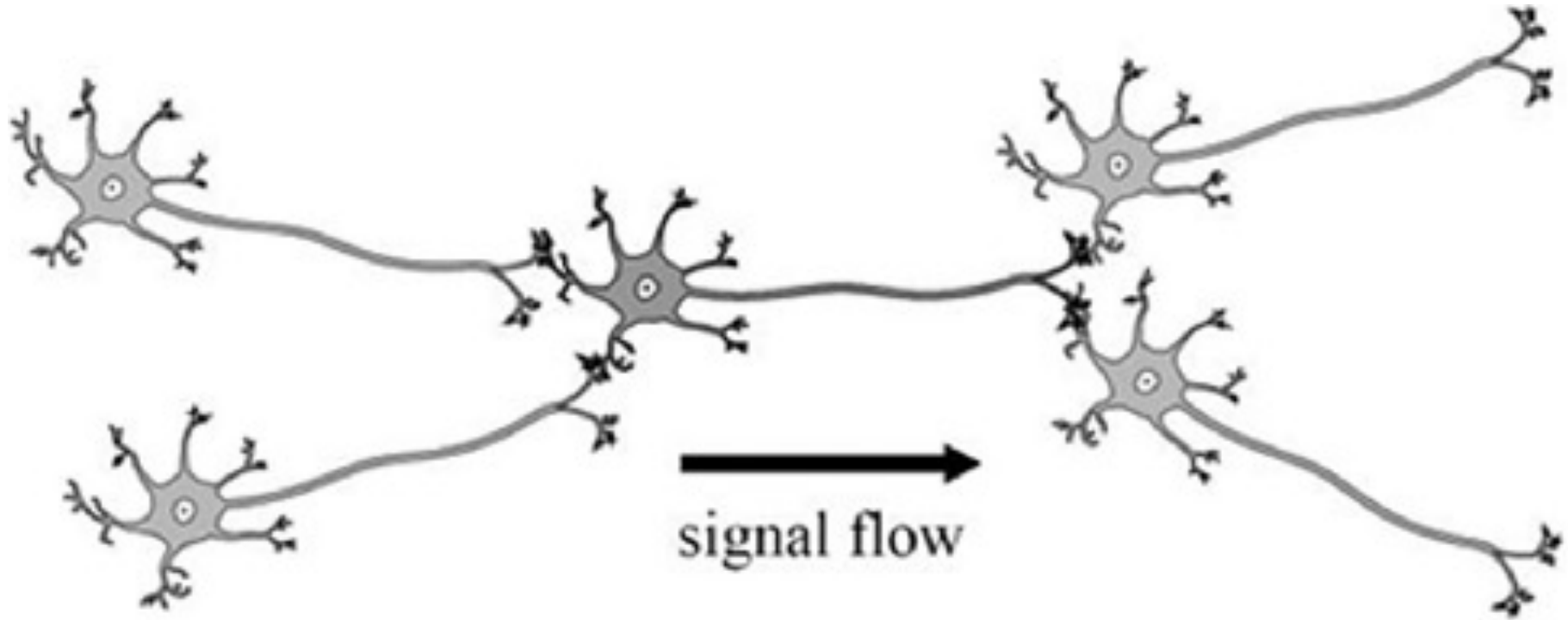
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Neurons



<https://www.visiblebody.com/learn/nervous/neurons>

Neural Networks



How can we study neural activities?